

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and developed **ETL pipelines** using **Python**, **SQL** (PostgreSQL, MySQL, Oracle), and **shell scripting** to extract, transform, and load financial transaction data across interconnected systems, ensuring **data integrity** and **99.5% SLA compliance** in production reconciliation workflows.
- Analyzed complex financial datasets to identify **data discrepancies** and implemented **data quality solutions** through automated validation scripts and ETL job monitoring using **Splunk** and **Dynatrace**, reducing manual reconciliation effort by **40%** and improving data accuracy across payment operations.
- Collaborated with business analysts, data engineers, and cross-functional stakeholders to gather reconciliation requirements, translate them into technical specifications, and deliver **data integration** solutions aligned with financial services regulatory requirements including **data governance**, **GDPR compliance**, and audit trail management.
- Monitored and troubleshooted **ETL jobs** in production environments, proactively identifying performance bottlenecks and failures using **CloudWatch** and log analysis; resolved **P1/P2 incidents** with **average resolution time of 2 hours**, ensuring uninterrupted data movement for reconciliation processes.
- Documented **ETL processes**, **data flows**, and **business rules** in **Confluence** to ensure transparency and knowledge sharing; created data lineage diagrams and technical runbooks for **change management** and UAT coordination across enterprise business systems and ERP modules.
- Contributed to **automation initiatives** using **Python** scripting, **Jenkins CI/CD pipelines**, and scheduled **cron jobs** to enhance efficiency in reconciliation workflows; built data pipeline orchestration solutions that reduced manual intervention by **60%** and improved **data warehousing** refresh cycles from daily to real-time ingestion.

SKILLS

ETL & Data Integration – ETL pipeline design and development, data extraction and transformation, ETL job monitoring, data flow architecture, data movement automation, ETL optimization.

Programming & Scripting – Python, SQL (PostgreSQL, MySQL, Oracle), shell scripting, data processing automation, cron job scheduling.

Data Management & Warehousing – data modeling, data warehousing concepts, data quality frameworks, data governance, relational databases, schema design, reconciliation processes, financial data management.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux, Git, Jenkins CI/CD pipelines, change-control management.

Monitoring & Observability – Splunk, Dynatrace, CloudWatch, SLA management, incident response, production support, performance troubleshooting.

Business Intelligence & Analytics – Power BI, Tableau, DAX, data modeling, business rules documentation, stakeholder reporting.

Financial Services Domain – payments reconciliation, SWIFT messaging (MT/MX), ISO 20022 message flows, SEPA, cross-border payments, PSD2, data governance in financial services, regulatory compliance, GDPR, audit trails.

EDUCATION AND TRAINING

PROJECTS

Reconciliation ETL Pipeline & Data Quality Framework (HCL Technologies)

- Architected and built an enterprise-grade **ETL pipeline** using **Python**, **SQL**, and **shell scripting** to support daily and real-time reconciliation of cross-border payment transactions, processing **2.5M+ records daily** with **99.95% data accuracy** across multiple source systems including ERP modules and SWIFT messaging platforms.
- Implemented automated **data quality checks** and discrepancy detection logic within the ETL flow using **pandas** and **NumPy**; designed **data validation rules** aligned with **ISO 20022 message structure** and reconciliation business logic, reducing manual investigation time by **55%** and enabling stakeholders to resolve exceptions **4x faster**.
- Integrated **ETL job monitoring** and alerting using **Splunk** and **CloudWatch**, capturing data lineage, transformation metrics, and pipeline performance; created operational dashboards using **Power BI DAX** measures to visualize **ETL SLA compliance**, **job failure rates**, and **data throughput** for business and operations teams.
- Established **data governance protocols** and **regulatory compliance** controls (GDPR, audit trail logging) within the pipeline; documented all **ETL processes**, **data flows**, and **business rules** in **Confluence** and created **change-management** runbooks for cross-functional UAT and post-go-live support.
- Orchestrated **ETL automation** using **Jenkins pipelines** and **cron-scheduled** batch jobs, eliminating **30+ hours/week** of manual reconciliation effort; implemented incremental load strategies and error-handling retry logic to achieve **99.5% on-time job completion** across production and non-production environments.

SWIFT Payments Reconciliation Dashboard & Operational Monitoring System (HCL Technologies)

- Led the design and deployment of a **reconciliation monitoring dashboard** for a banking client's SWIFT messaging operations, integrating **real-time transaction data** from payment platforms, ERP systems, and SWIFT MT/MX message logs using **SQL queries** and **ETL data pipelines**.
- Built interactive **Power BI dashboards** with **DAX measures** and **data modeling** to surface transaction-flow visibility, settlement status, and **reconciliation discrepancies** across **SEPA**, **cross-border payments**, and **ISO 20022** message types, enabling operations teams to detect anomalies and resolve exceptions **within SLA**.
- Collaborated with banking domain business analysts and regulatory stakeholders to embed **data governance**, **audit trail logging**, and **GDPR compliance** controls into the operational reporting layer, ensuring **regulatory requirements** and **financial services data protection** standards were met across all reconciliation workflows.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)